

```
import random
```

```
file = open("student_performance.csv", "w")
```

```
file.write("math_score,reading_score,writing_score,placement_score,club_join_date\n")
```

```
70
```

```
for i in range(45):
    math_score = random.randint(60, 80)
    reading_score = random.randint(75, 95)
    writing_score = random.randint(60, 80)
    placement_score = random.randint(75, 100)
    club_join_date = random.randint(2020, 2025)

    row = f"{math_score},{reading_score},{writing_score},{placement_score},{club_join_date}\n"
    file.write(row)
```

```
file.close()
```

```
print("student_performance.csv created successfully!")
```

```
student_performance.csv created successfully!
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

```
df = pd.read_csv("student_performance.csv")
```

```
print(df.head())
```

	math_score	reading_score	writing_score	placement_score	club_join_date
0	62	89	74	99	2024
1	70	80	69	85	2022
2	63	88	72	98	2025
3	75	80	73	85	2021
4	75	92	60	94	2022

```
print(df.isnull().sum())
```

```
math_score      0
reading_score    0
writing_score    0
placement_score  0
club_join_date   0
dtype: int64
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df.dropna(inplace=True)
```

```
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1

outliers = ((df < (Q1 - 1.5 * IQR)) | (df > (Q3 + 1.5 * IQR)))
print(outliers.sum())
```

```
math_score      0
reading_score    0
writing_score    0
placement_score  0
club_join_date   0
dtype: int64
```

```
df = df[~((df < (Q1 - 1.5 * IQR)) |
          (df > (Q3 + 1.5 * IQR))).any(axis=1)]
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

score_columns = ["math_score", "reading_score", "writing_score", "placement_score"]

df[score_columns] = scaler.fit_transform(df[score_columns])

print(df.head())
```

	math_score	reading_score	writing_score	placement_score	club_join_date
0	0.10	0.722222	0.70	0.96	2024
1	0.50	0.222222	0.45	0.40	2022
2	0.15	0.666667	0.60	0.92	2025
3	0.75	0.222222	0.65	0.40	2021
4	0.75	0.888889	0.00	0.76	2022

```
print(df[score_columns].skew())
```

```
math_score      -0.293744
reading_score    0.231936
writing_score    -0.185111
placement_score -0.213141
dtype: float64
```

```
for col in score_columns:
    df[col] = np.log1p(df[col])
```

```
df[score_columns].hist(figsize=(8,6))
plt.show()
```



